

Aula Conectada

Investment case · Bolivia

Hernando Grueso, PhD
UNICEF

2026-05-01

1 · Diagnostic — gaps in education

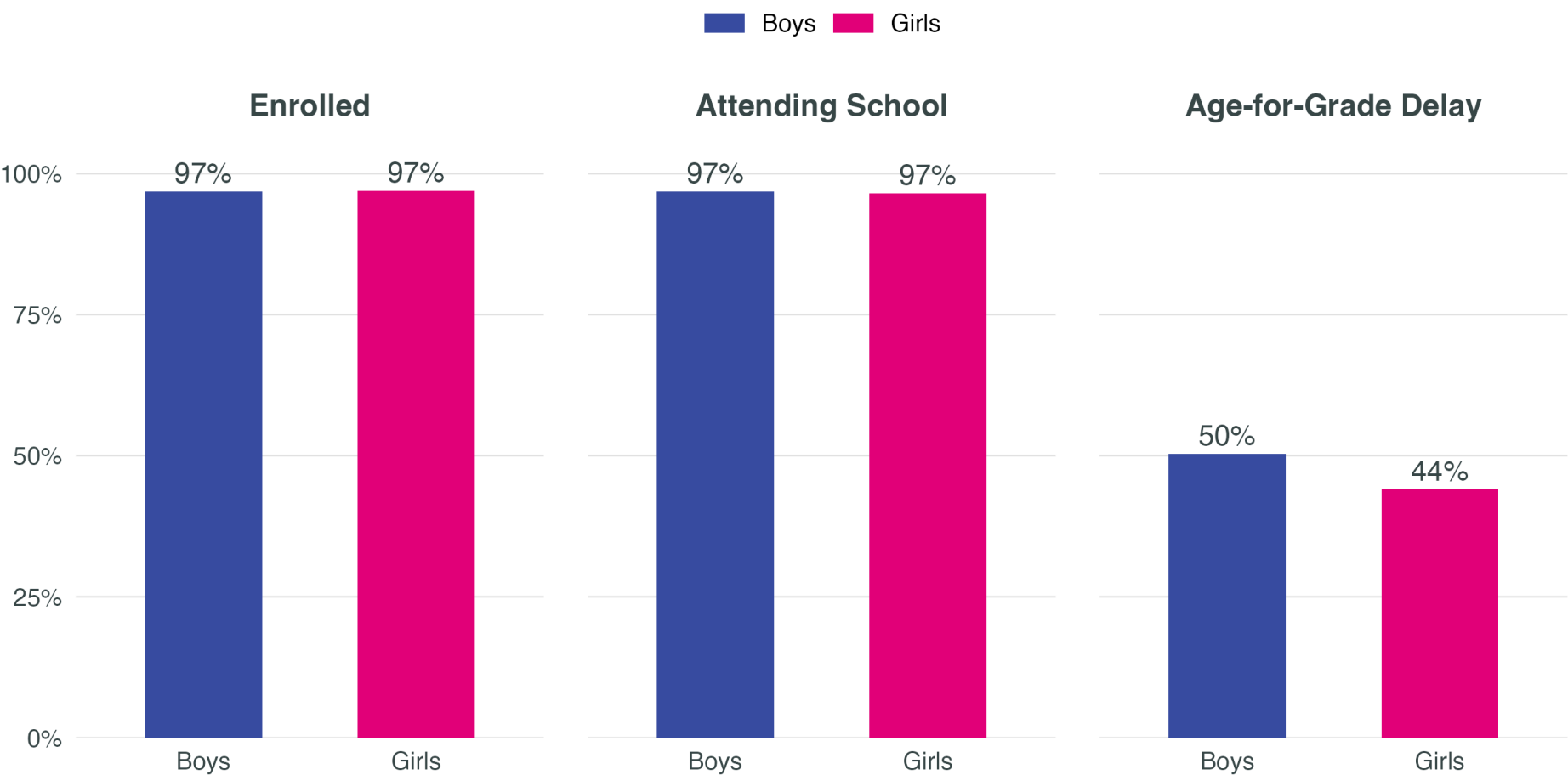
Target population — adolescent girls 10–17 in Bolivia

Segment	Weighted N	% of girls 10–17
All girls 10–17	976,569	100.0%
Rural girls 10–17	281,797	28.9%
Indigenous girls 10–17	146,455	15.0%
Girls in poverty (INE) 10–17	437,692	44.8%
Girls with disability (WG 3+) 10–17	4,142	0.4%
Girls without HH internet 10–17	215,766	22.1%
Priority girls (rural ∨ Indigenous ∨ poor ∨ disability)	567,836	58.1%

Gender gaps in education

Gender gaps in education outcomes

Adolescents 10–17 in Bolivia

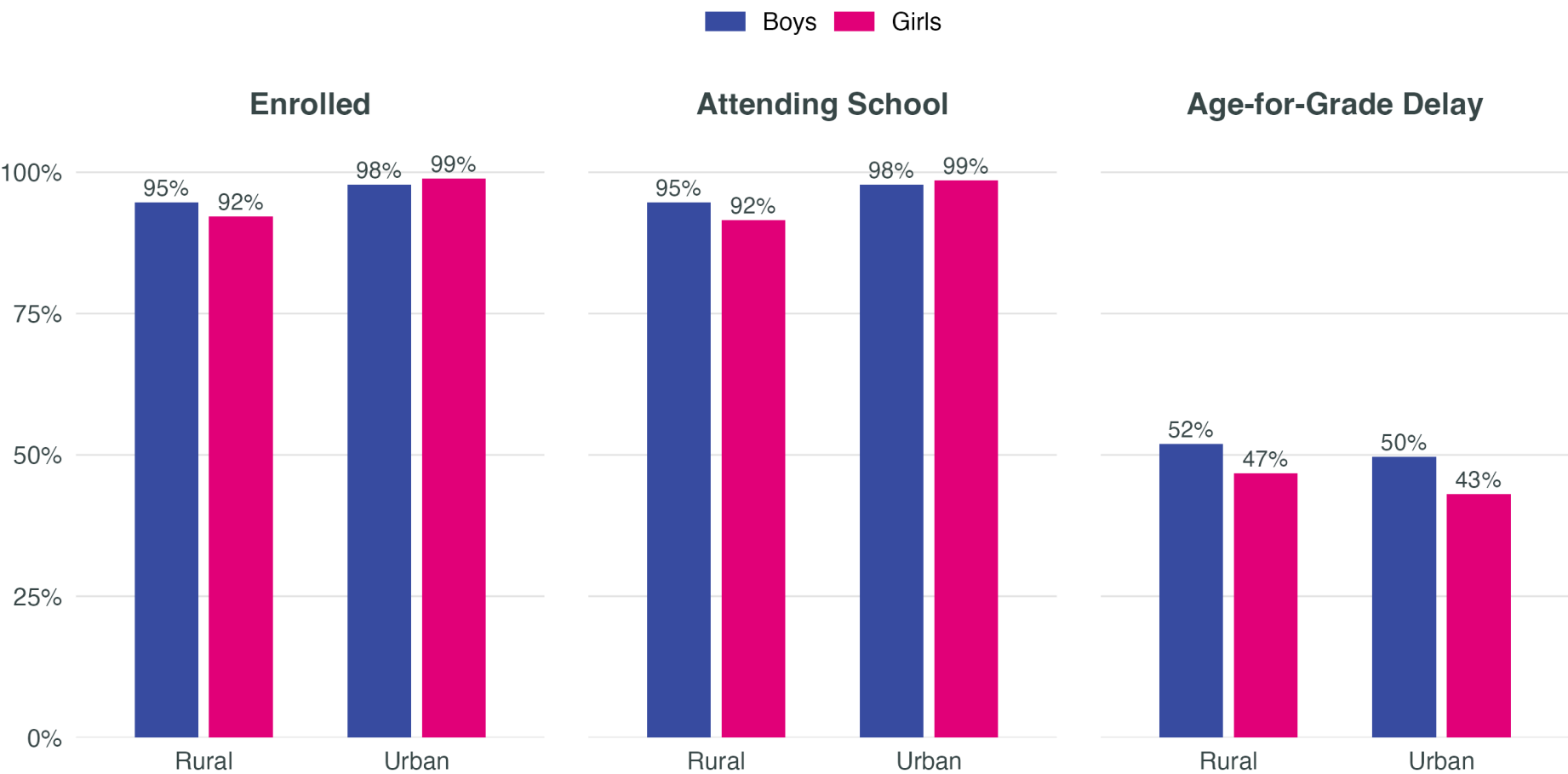


Source: INE Bolivia – Household Survey 2024. Weighted estimates, adolescents 10–17.

Rurality amplifies education gaps

Education outcomes by area of residence

Adolescents 10–17, by sex and rural/urban

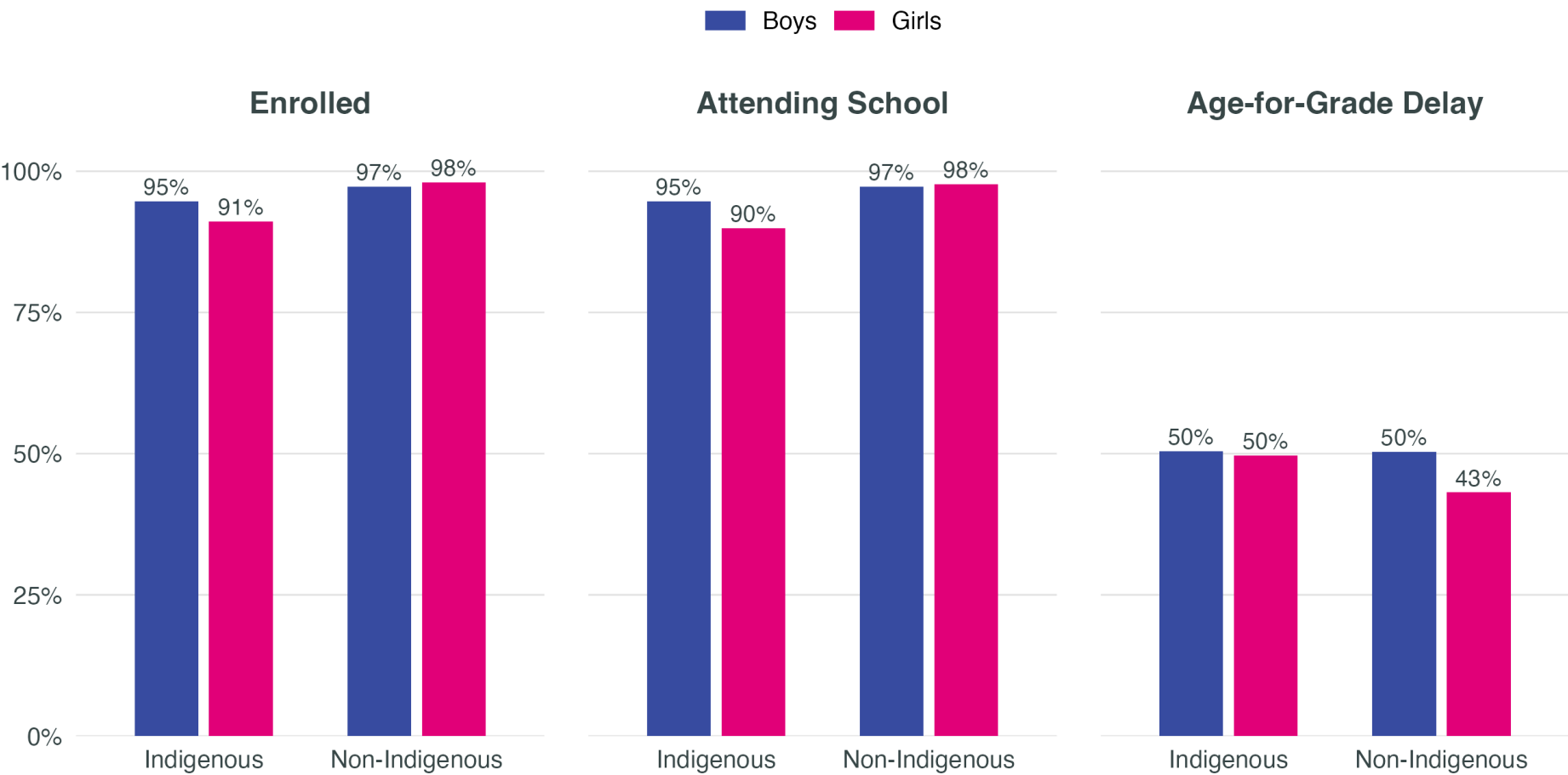


Source: INE Bolivia – Household Survey 2024. Weighted estimates, adolescents 10–17.

Indigenous girls face differential grade delay

Education outcomes by Indigenous heritage

Adolescents 10–17, by sex

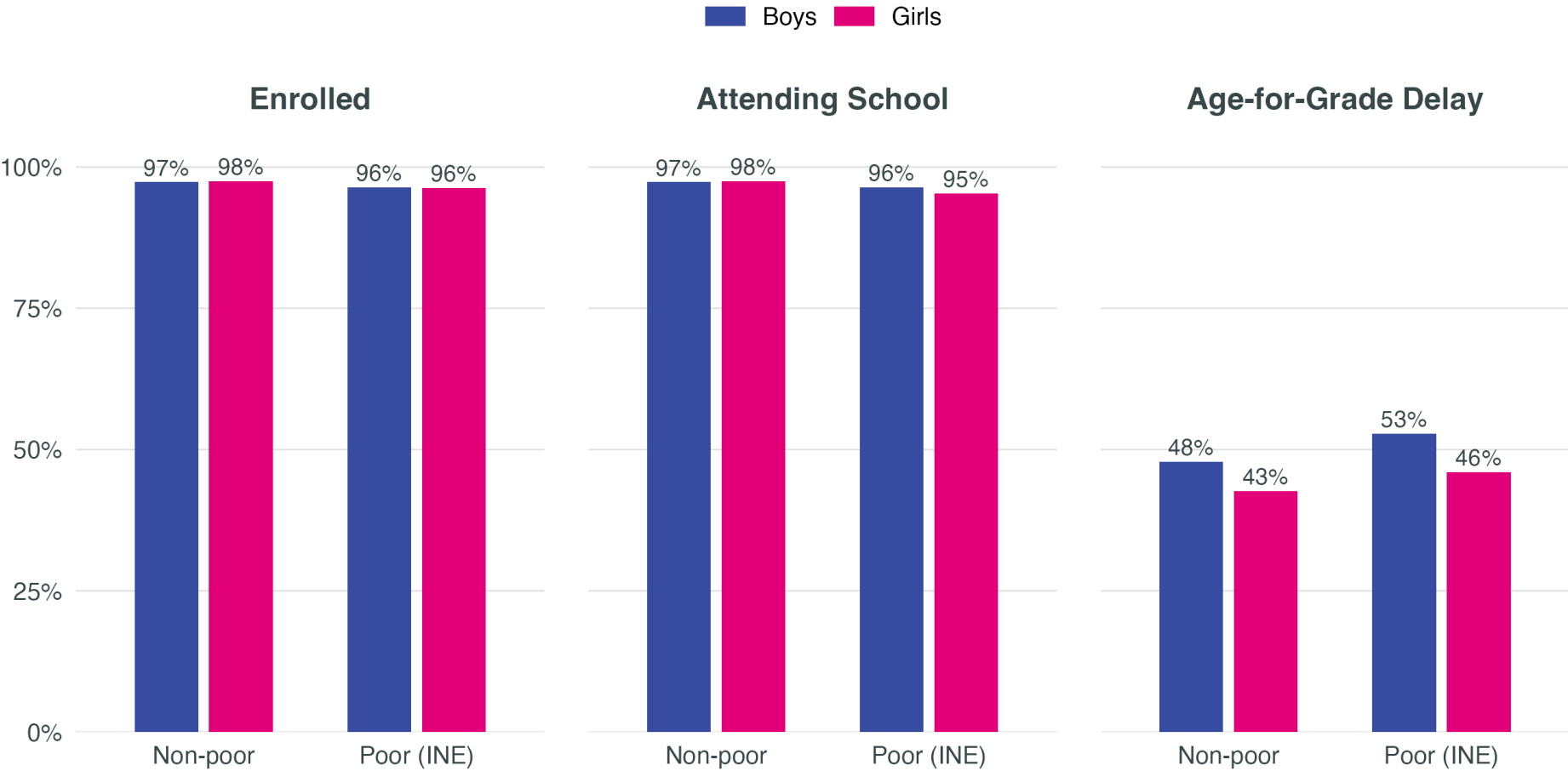


Source: INE Bolivia – Household Survey 2024. Weighted estimates, adolescents 10–17.

Poverty compounds education gaps

Education outcomes by poverty status

Adolescents 10–17, by sex (INE)

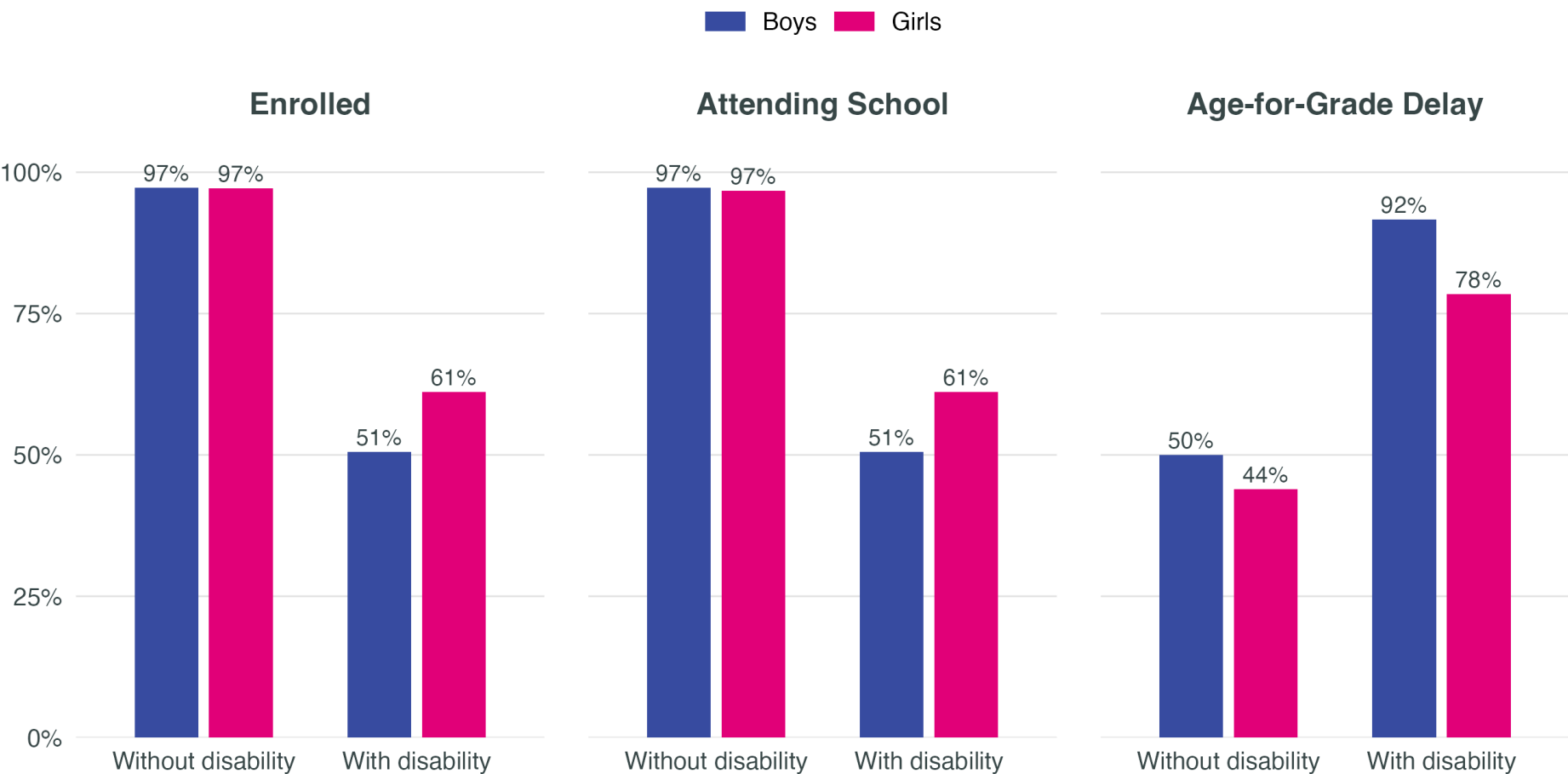


Source: INE Bolivia – Household Survey 2024. Weighted estimates, adolescents 10–17.

Girls with disabilities — compounded exclusion

Education outcomes by disability status

Adolescents 10–17, by sex · Washington Group severity ≥ 3



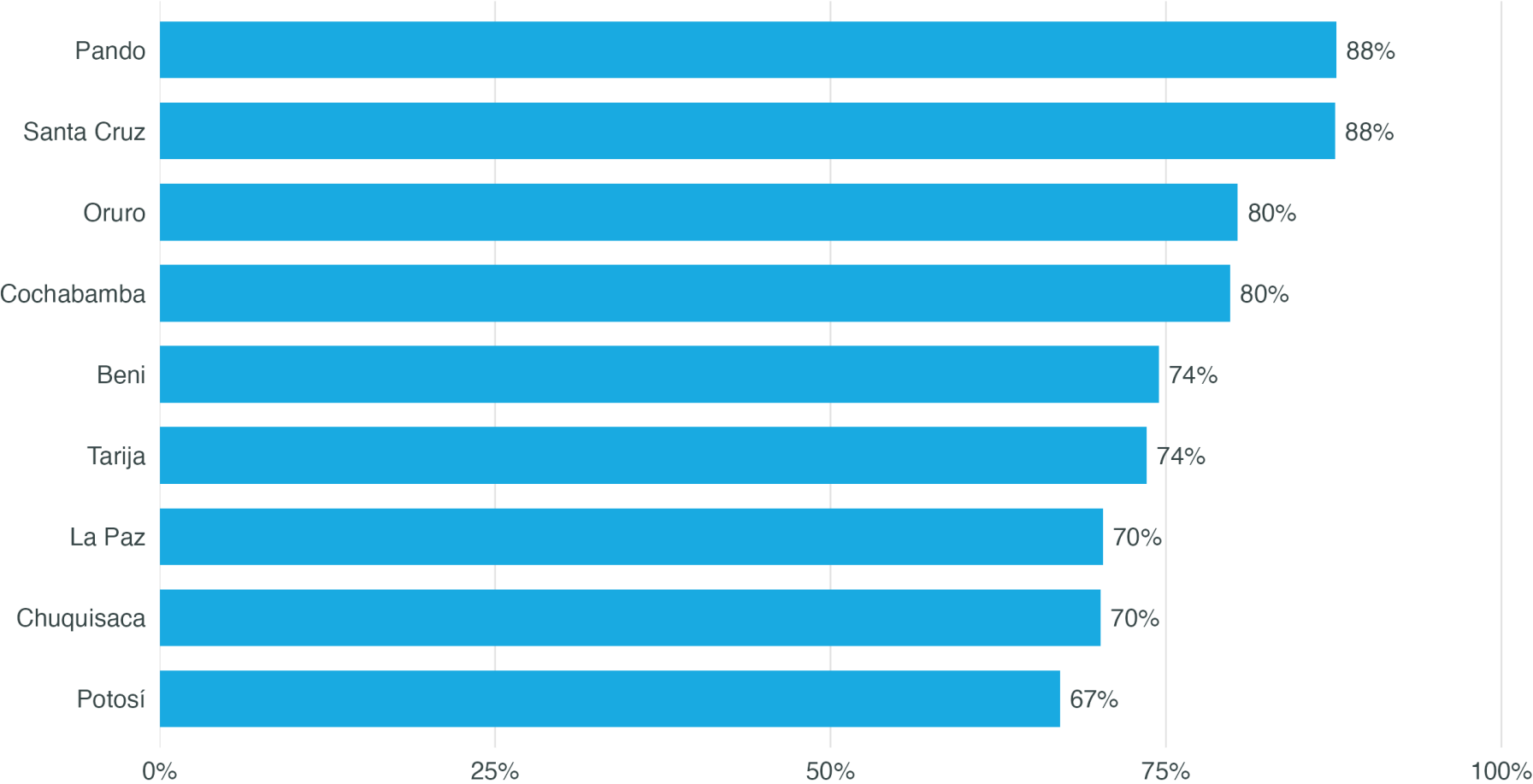
Source: INE Bolivia – Household Survey 2024. Weighted estimates, adolescents 10–17. Small sample (~0.4%) — interpret with caution.

2 · Why digital — the mechanisms

The Digital Divide

Territorial variation — input for prioritisation and sequencing

HH internet, adolescent girls 10–17, by department

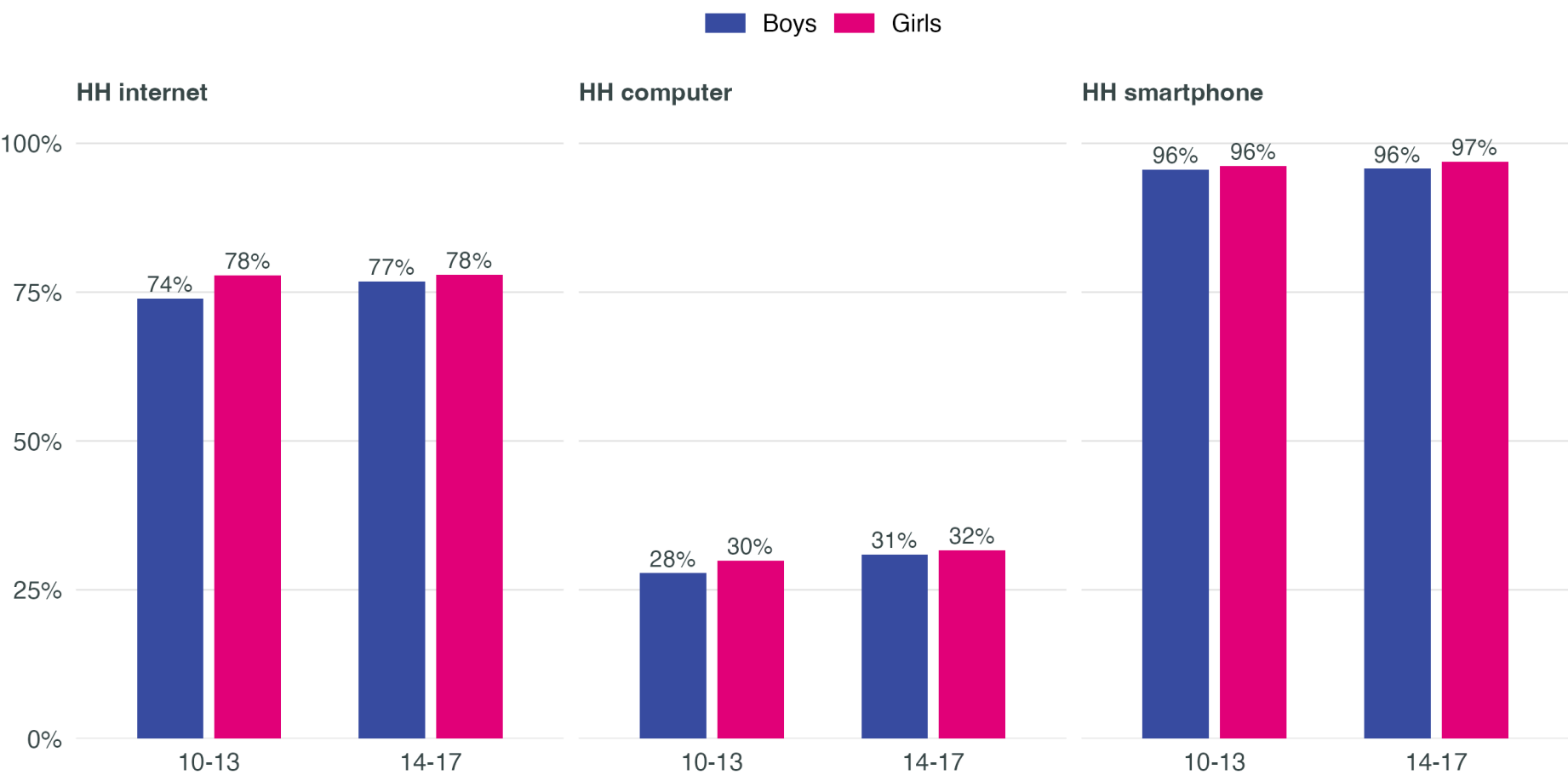


Source: INE Bolivia – Household Survey 2024. Weighted estimates, adolescents 10–17.

Household digital access — the material enabler

Household digital access for adolescents in Bolivia

Share of girls and boys in households with each digital asset, by age group

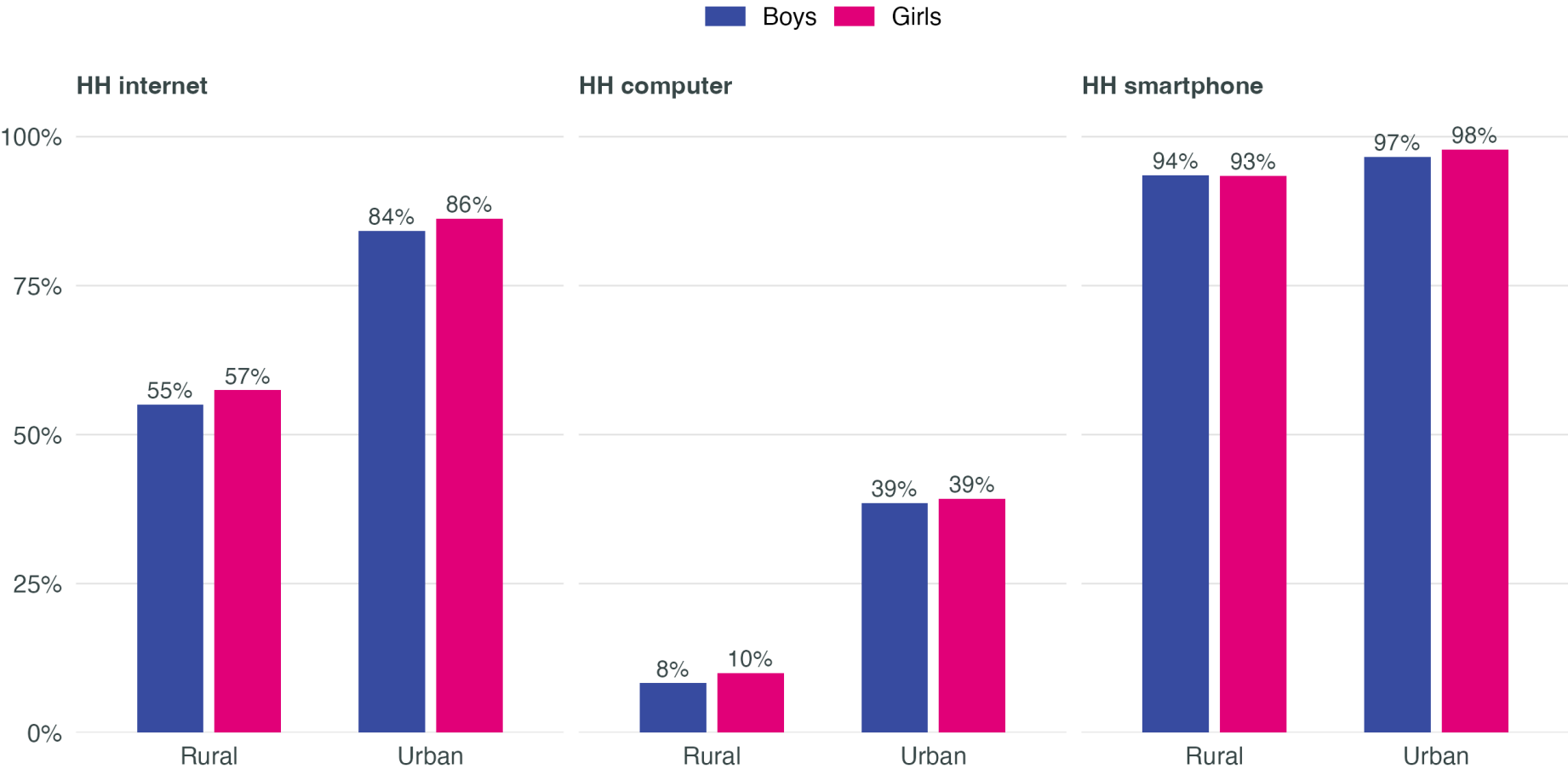


Source: INE Bolivia – Household Survey 2024. Weighted estimates, adolescents 10–17.

The largest digital divide is territorial, not just gender

The largest digital divide is territorial, not just gender

Adolescents 10–17, by sex and area

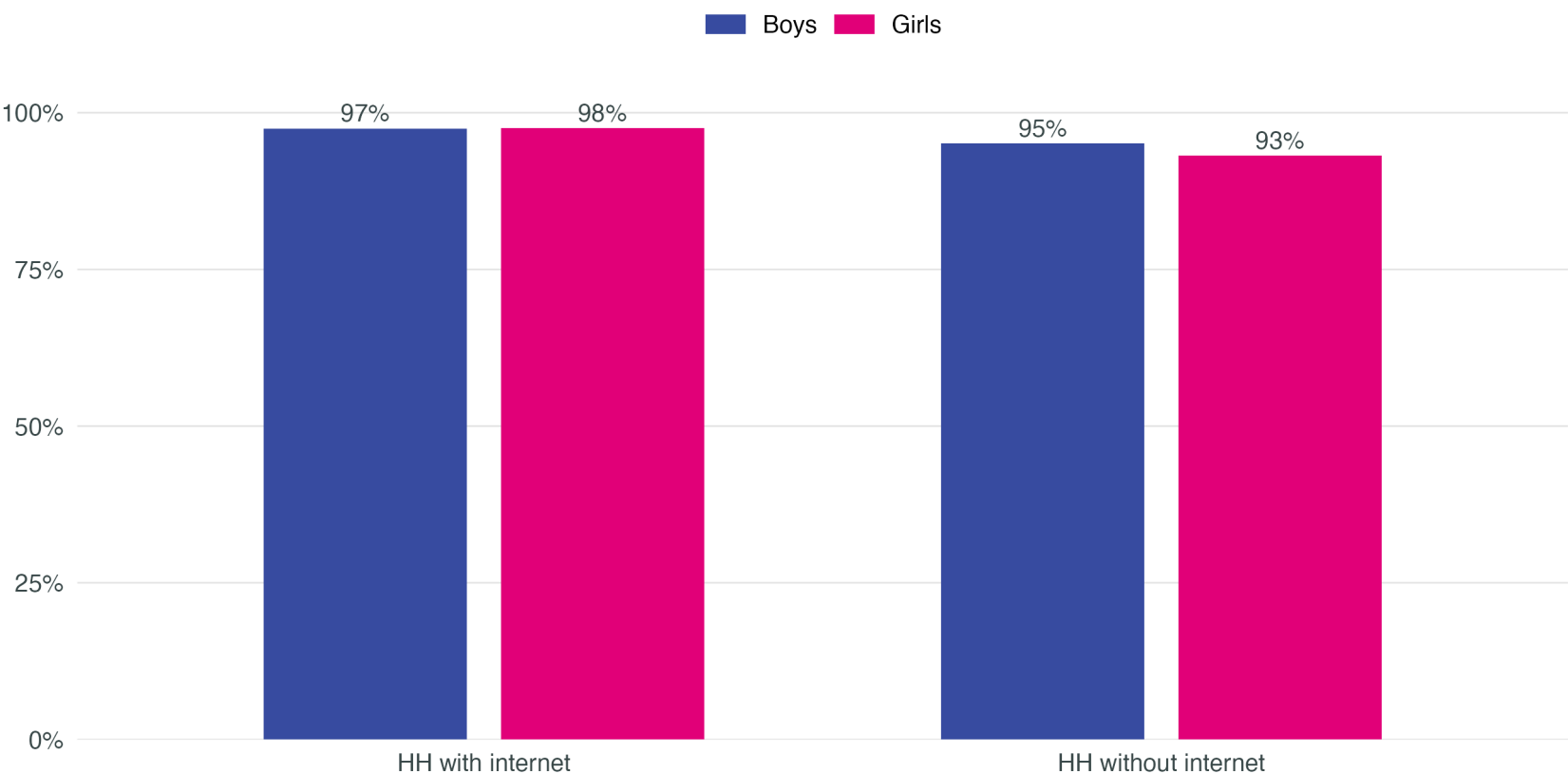


Source: INE Bolivia – Household Survey 2024. Weighted estimates, adolescents 10–17.

Connected households show higher school attendance for girls

Connected households show higher school attendance

Share of adolescents 10–17 currently attending school



Source: INE Bolivia – Household Survey 2024. Weighted estimates, adolescents 10–17.

3 · Multivariate analysis

Controlled gender gaps — main coefficients

Predictor	Enrolled	Attending	Grade delay	HH internet	Any HH device
Girl (1=yes)	-0.1 pp	-0.5 pp	-6.0 pp***	+1.4 pp	+0.4 pp
Rural	-4.4 pp**	-4.4 pp**	+3.2 pp	-23.2 pp***	-4.5 pp**
Indigenous	-3.4 pp**	-3.7 pp**	+3.2 pp	-1.7 pp	-0.7 pp
Disability (WG3+)	-43.8 pp***	-43.6 pp***	+40.3 pp***	-0.9 pp	+3.7 pp***
Poor (INE)	-0.1 pp	-0.6 pp	+5.6 pp***	-14.7 pp***	-1.7 pp*

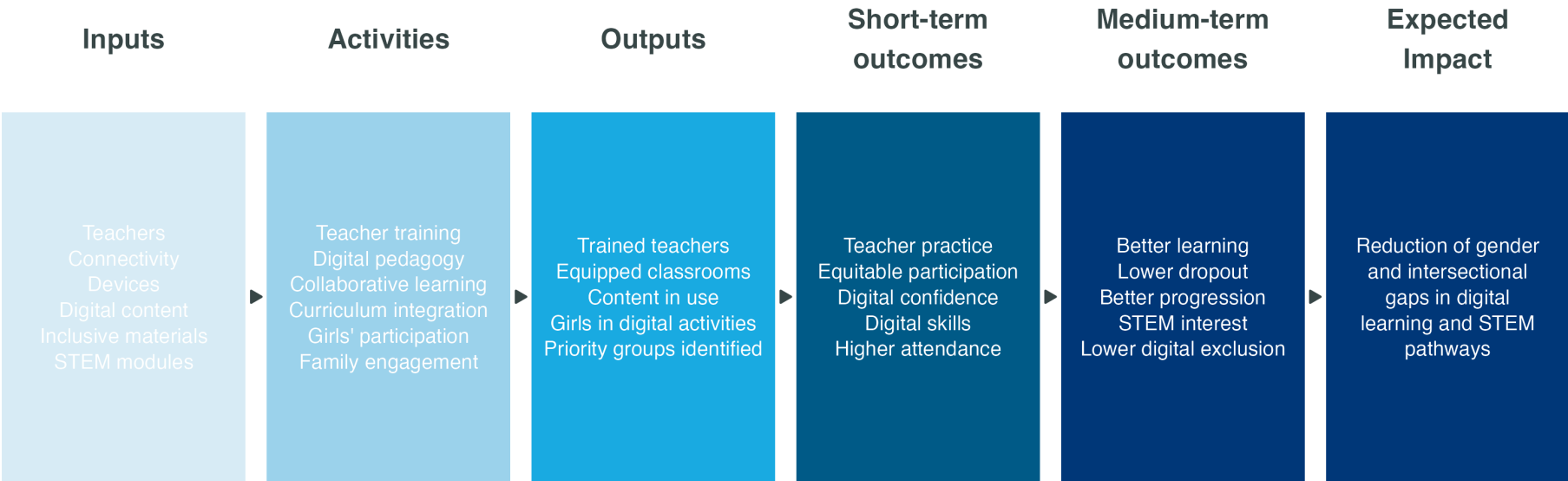
Key reading: girl coefficient is not significantly associated with average gaps in enrollment or attendance, but is **associated with lower grade delay (-6 pp***)** — girls who are in school progress better. The strong gaps are **rural, poverty, and Indigenous**. Weighted OLS with department fixed effects; PSU-clustered SE. Details and intersections in appendix (A3–A4).

4 · Theory of Change

From inputs to impact — proposed Theory of Change

Theory of Change — Aula Conectada

From inputs to impact: technology as means, pedagogy as engine



Programme components and expected outcomes

Core components

- Teacher training in digital pedagogy and gender-responsive teaching
- Structured digital content integrated into the curriculum
- Collaborative learning
- Inclusion adaptations (disability, language, rurality)
- Girls' active participation in STEM
- **Family engagement** — parenting layer that reduces restrictive gender norms and intra-household violence

Expected outcomes

- Improved teacher practice
- Greater girls' participation in class
- Higher digital confidence
- Better digital skills
- Improved attendance and progression
- Higher STEM interest
- **Lower violence against girls in the home** (via parenting layer)

What the international evidence tells us

Consistent findings

- Hardware alone rarely improves learning
(De Melo et al., 2014; Cristia et al., 2017)
- Digital tools work when embedded in structured pedagogy
(Gómez-Ruiz & Franco, 2023)
- Teacher training is the central delivery mechanism
(Olugbade et al., 2025; Bassi et al., 2016)
- Gender-responsive content improves STEM aspirations
(Outes-León et al., 2020; Porter & Trzesniewski, 2020)

Parenting layer (new)

- Parenting interventions reduce violence against girls
(Cluver et al., 2017 — South Africa)
- Digital parenting apps improve practices
(Janowski et al., 2025 — Tanzania)
- Restrictive gender norms at home are a structural barrier to learning

5 · Methodology — adapting evidence to the Bolivian context

How we adjust each external anchor

Each external anchor is adjusted using **five factors** derived solely from observable variables.

Adjusted effect = external effect $\times \phi_{\text{pop}} \times \phi_{\text{baseline}} \times \phi_{\text{geo}} \times \phi_{\text{econ}} \times \phi_{\text{delivery}}$

- **ϕ_{pop}** — population similarity (age, % rural)
- **ϕ_{baseline}** — headroom given Bolivia's current level
- **ϕ_{geo}** — geographic distance from La Paz
- **ϕ_{econ}** — GDP per capita ratio Bolivia/anchor
- **ϕ_{delivery}** — government (1.0) vs research/NGO (0.7)
- **95% CI** — propagated from the anchor's reported SE

Anchors not projected: age range outside 10–17 (ProJoven, Irûmi) or insufficient demographic data. Full methodology in A6.

Structural Equation Model — synthesising the ToC

The SEM integrates all anchors into a single model of the Theory of Change, calibrated with ϕ -adjusted parameters as informative priors.

Key structural arrows

- Teacher training → Teacher practice
- Teacher practice → Girls' digital skills
- **Parenting → Home environment (norms, violence) → Attendance**
- Attendance → Progression → Learning
- Learning → Long-term labor trajectories

Calibration

- Learning parameters: digital-education anchors
- Parenting parameters: Cluver + Janowski
- Home-environment outcomes: **MICS Bolivia 2019** (next phase)
- Bayesian estimation with **blavaan** to propagate CI

Pre-registration of equations in A9. Estimation pending — requires MICS + costing.

6 · Anchors and projections

Direct anchors — digital education and gender norms

Programmes with measured effects on the educational outcomes Aula Conectada targets:

Programme	Country	External effect (95% CI)	ϕ_{pop}	ϕ_{geo}	ϕ_{econ}	$\phi_{\text{deliv.}}$	n par.	Adjusted effect (95% CI)	Sig.
Mentalidad de crecimiento (Outes & Sanchez, full)	Perú	0.054 SD [-0.005, 0.113]	0.48	0.85	0.50	1.00	4	0.011 SD [-0.001, 0.023]	—
Mentalidad de crecimiento (Outes & Sanchez, regional)	Perú	0.135 SD [0.035, 0.235]	0.48	0.85	0.50	1.00	4	0.028 SD [0.007, 0.048]	✓
Mentalidad de crecimiento (Porter et al.)	Sudáfrica	-0.040 SD [-0.236, 0.156]	0.46	0.50	0.54	0.70	4	-0.003 SD [-0.020, 0.014]	—
ELA (Bandiera et al., 12-17 cohort)	Sierra Leona	7.2 pts [2.0, 12.5]	0.45	0.50	1.00	0.70	5	1.1 pts [0.3, 2.0]	✓

Reading the CI: if zero falls inside the interval, the anchor's original effect is not statistically distinguishable from zero — this reflects the study's uncertainty, not an error in the adjustment. The **Sig.** column marks ✓ when the adjusted effect is significant. **ELA caveat:** high population similarity reflects observable overlap (rural, age), but Sierra Leone's post-Ebola context differs substantially.

Indirect anchors — parenting layer (via SEM)

Parenting programmes that **do not act directly** on enrollment/attendance, but on the home environment (norms, violence). They enter the SEM as parameters for the parenting → home environment → education-outcome arrow.

Programme	Country	Original outcome	ϕ_{pop}	ϕ_{geo}	ϕ_{econ}	$\phi_{deliv.}$	Use in model
ParentApp Tanzania (Janowski 2025)	Tanzania	see A9 (SEM)	0.24	0.50	1.00	0.70	parenting → home environment input parameter
Parenting for Lifelong Health (Cluver et al. 2017)	Sudáfrica	see A9 (SEM)	1.00	0.50	0.54	0.70	parenting → home environment input parameter

Cluver et al. (2017) reports effects on physical and emotional abuse (IRR 0.55, not directly translatable to enrollment). Janowski et al. (2025) reports engagement with the app (modules completed). Effects propagate to educational outcomes via the SEM, not directly.

Key assumptions

Each projection rests on explicit assumptions. Documenting them allows the Ministry to evaluate the credibility of each estimate.

On the external anchors

- Studies with credible identification in their original context (RCT or quasi-experimental)
- **Causal mechanisms** partially transferable (teacher training, structured pedagogy, gender-responsive content operate comparably across countries)
- The observable variables we use capture the relevant heterogeneity

On Bolivia and implementation

- Aula Conectada will be implemented with **pedagogical fidelity** comparable to the anchors
- Gaps observed in EH 2024 are representative of the scale-up context
- Priority subgroups will remain the target population

How assumptions are handled: each ϕ factor is computed on observable data. When data are insufficient (e.g., ϕ_{baseline} without reported baseline), the factor is explicitly set to 1.0. The **n par.** column indicates how many factors were calibrated with real data.

Synthesis — expected returns from Aula Conectada

Key lesson from Plan Ceibal (De Melo et al., 2014): the hardware-only version had zero effect on learning. **Aula Conectada captures returns only if it funds the full pedagogical package**, not just equipment.

Short-to-medium term

(priority group, ≈ 568K girls)

- **Learning (math):** +0.09 SD ↔ +0.18 SD

(Outes-León et al., 2020; Porter & Trzesniewski, 2020)

- **Computational thinking and STEM:** +0.06 SD to +0.08 SD

(Näslund-Hadley et al., 2025)

- **Prevented dropout** in higher-exclusion areas: 7–14 pp

(Bandiera et al., 2018)

Long-term returns

(comparable regional evidence)

- **+15.2 pp** in female employment rate

(Ñopo et al., 2008 — ProJoven Peru)

- **+93%** in female labor income

- **Mechanism:** early digital skills → school progression → employability

7 · Costing data

Cost per girl — pending

8 · Cost-effectiveness analysis

Cost-effectiveness of Aula Conectada — pending

9 · Methodological Appendix

A1 · Data and sample

Source: Encuesta de Hogares 2024 (EH 2024) — INE Bolivia · Collection November–December 2024 · National coverage, urban and rural · Unit of analysis: household members in private dwellings.

Sample: 12,718 households · 39,497 persons · Analytical subsample: **6,254 adolescents 10–17** (≈ 977K girls and 1.05M boys expanded) · Files: Persona, Vivienda, Equipamiento.

Key variables

- **Girl** = `s01a_02 == 2` · **Rurality** = `area == 2` · **Poverty** = `p0` (INE)
- **Indigenous heritage** = mother tongue or spoken language in Bolivian codes (1–37, 70, 71)
- **Disability** = Washington Group, severity ≥ 3 on any domain (vision, hearing, mobility, memory)
- **HH internet** = `s06a_19` · **Digital equipment** (computer, smartphone, tablet) from Equipamiento module

Estimation: weighted OLS using `factor` (sampling weight); SE clustered at PSU (`fixest::feols`).

A2 · Empirical strategy

Base model (LPM with department fixed effects):

$$Y_{ihd} = \alpha + \beta_F F_i + \beta_R R_i + \beta_I I_i + \beta_D D_i + \beta_P P_i + \gamma X_{ihd} + \delta_d + \varepsilon_{ihd}$$

where F = Girl, R = Rural, I = Indigenous, D = Disability, P = Poor, X = controls (age).

Intersectional model with two-way interactions (× Girl):

$$Y_{ihd} = \alpha + \beta_F F_i + \gamma X_{ihd} + \sum_{k \in \{R, I, D, P\}} \theta_k (F_i \times k_i) + \delta_d + \varepsilon_{ihd}$$

Intersectional model with three-way interactions (robustness):

$$Y_{ihd} = \alpha + \text{main effects} + 2\text{-way} + \theta_{F \cdot R \cdot I} (F_i \times R_i \times I_i) + \delta_d + \varepsilon_{ihd}$$

and analogues for $F \times R \times P$ and $F \times I \times P$.

Weighted OLS using **factor** (sampling weight); standard errors clustered at PSU level (**fixest::feols**).

A3 · Full table — main coefficients

Variable	HH internet	Enrolled	Attending	Any HH device	Grade delay
Girl (1=yes)	0.014	-0.001	-0.005	0.004	-0.060***
	(0.013)	(0.006)	(0.006)	(0.005)	(0.015)
Rural	-0.232***	-0.044**	-0.044**	-0.045**	0.032
	(0.042)	(0.017)	(0.018)	(0.020)	(0.022)
Poor (INE)	-0.147***	-0.001	-0.006	-0.017*	0.056***
	(0.023)	(0.007)	(0.008)	(0.009)	(0.018)
Indigenous	-0.017	-0.034**	-0.037**	-0.007	0.032
	(0.039)	(0.014)	(0.014)	(0.017)	(0.027)
Disability (WG3+)	-0.009	-0.438***	-0.436***	0.037***	0.403***
	(0.080)	(0.080)	(0.080)	(0.012)	(0.055)
N (observations)	6,254	6,254	6,254	6,254	6,254
R ²	0.148	0.081	0.078	0.036	0.033
Department FE	Yes	Yes	Yes	Yes	Yes
Sample weights	Yes	Yes	Yes	Yes	Yes
PSU clustered SE	Yes	Yes	Yes	Yes	Yes

* p<0.10, ** p<0.05, *** p<0.01. SE in parentheses below each coefficient. Weighted OLS using EH 2024 expansion factor; department fixed effects; PSU-clustered errors (`fixest::feols`).

A4 · Full intersectional table

Interaction	Enrolled	Attending school	HH internet	Any HH device
Girl × Rural	-0.037**	-0.040**	0.010	-0.011
	(0.017)	(0.018)	(0.035)	(0.013)
Girl × Indigenous	-0.040**	-0.049**	-0.004	-0.012
	(0.018)	(0.019)	(0.046)	(0.015)
Girl × Disability	0.106	0.111	-0.112	-0.018
	(0.178)	(0.178)	(0.150)	(0.027)
Girl × Poor	-0.004	-0.013	0.038	-0.013
	(0.012)	(0.014)	(0.033)	(0.013)
N (observations)	6,254	6,254	6,254	6,254
R ²	0.083	0.081	0.148	0.037
Department FE	Yes	Yes	Yes	Yes
Sample weights	Yes	Yes	Yes	Yes
PSU clustered SE	Yes	Yes	Yes	Yes

Interaction coefficients (additional effect for girls in each group). SE in parentheses. * p<0.10, ** p<0.05, *** p<0.01. Weighted OLS, department fixed effects, PSU-clustered errors.

A5 · Heterogeneity analysis — regression forest

Method: `grf::regression_forest` (Athey & Wager, 2019) — ensemble of regression trees with honest sub-sampling.

Specification

- **Training sample:** all adolescents 10–17 (not only girls) — lets the forest learn the gender effect endogenously
- **Covariates (7):** sex, age, rurality, Indigenous, disability, poverty, department
- **Outcomes (4 separate forests):** attendance, HH internet, any device, computer
- **Hyperparameters:** 2,000 trees · honest fitting · fraction 0.5 · sample weights = EH 2024 expansion factor

What we report

- **Variable importance** normalised per outcome
- **Subgroup predictions** on a 16-profile grid (sex × rural × Indigenous × poverty), fixing age = 13 and department = La Paz

What this is NOT

- Not a causal forest (no treatment)
- Does not estimate the effect of Aula Conectada
- Estimates **baseline** gap heterogeneity — useful for targeting

A6 · Bolivia-adjusted projection methodology

Full formula: $\text{Effect} = \beta_{\text{ext}} \times \phi_{\text{pop}} \times \phi_{\text{baseline}} \times \phi_{\text{geo}} \times \phi_{\text{econ}} \times \phi_{\text{delivery}}$

ϕ_{pop} — $\exp(-\bar{d}^2)$ over age and % rural, standardised. Indig_share **deliberately excluded** (not apples-to-apples across contexts). Bounded [0,1].

ϕ_{baseline} — $(Y_{\text{max}} - Y_{\text{Bol}})/(Y_{\text{max}} - Y_{\text{anchor}})$. NA if anchor reports no baseline. Bounded [0,1].

ϕ_{geo} — Haversine distance from La Paz. Same country = 1.00; <3000 km = 0.85; <6000 km = 0.65; ≥ 6000 km = 0.50.

ϕ_{econ} — GDP per capita ratio Bolivia/anchor. If anchor is from a poorer country $\rightarrow 1.0$ (no haircut). Bounded [0.3, 1].

ϕ_{delivery} — 1.00 if anchor is government-implemented, 0.70 if research/NGO (Vivalt 2020).

Exclusion rules — age overlap <25% with 10–17, or <2 demographic variables comparable $\rightarrow$ anchor not projected.

Uncertainty — 95% CI propagated from the anchor's SE. No synthetic noise on ϕ factors.

A7 · Formal foundation of the transportability approach

We assume the mismatch between the source population S (the anchor study) and the target population T (Bolivia) decomposes into K observable dimensions, with **multiplicative** attenuation:

$$\hat{\beta}_T \approx \hat{\beta}_S \cdot \prod_{k=1}^K \phi_k(S, T), \quad \phi_k \in [0, 1]$$

where $\hat{\beta}_S$ is the effect reported in the anchor study, $\hat{\beta}_T$ the projected effect for Bolivia, ϕ_k the adjustment factor for dimension k , and $K = 5$ the mismatch dimensions (population, baseline, geography, economy, delivery mode).

Each factor is bounded in [0, 1]:

- ϕ_{pop} — standardised age and % rural
- ϕ_{baseline} — initial outcome level
- ϕ_{geo} — Haversine distance from La Paz
- ϕ_{econ} — GDP per capita ratio
- $\phi_{\text{delivery}} = 1.0$ (government) or 0.7 (NGO/research)

Uncertainty propagation. The 95% CI is propagated from the anchor's SE:

$$\text{SE}(\hat{\beta}_T) = \text{SE}(\hat{\beta}_S) \cdot \prod_{k=1}^K \phi_k$$

Key assumptions (claims, not estimates):

- (A1) **Sufficiency of observables:** the 5 dimensions capture the relevant effect modifiers between S and T
- (A2) **Separability:** each ϕ_k effect is approximately multiplicative
- (A3) **Monotonicity:** mismatch only attenuates, never amplifies
- (A4) **SE independence:** anchor variance propagates without perturbing the ϕ_k

A8 · Consolidated evidence — anchors in use

Paper	Country	Design	Age	Effect	ToC arrow
Näslund-Hadley_2025 · Irumi Program Paraguay	PRY	RCT	n.r.	0.089 (0.052)	Digital education
Bandiera_2018 · ELA Sierra Leone Ebola Context	SLE	RCT	12–25	0.085 (0.049)	Gender norms
Outes-León_2020 · Power of Believing Peru	PER	RCT	n.r.	0.054 (0.030)	Other

Only papers used as anchors in the Bolivia projection are shown (see [05_projections.R](#) and [R/anchors_demographics.R](#)). The full systematic review (52 papers) is in [data/lit_review_papers.csv](#). The **ToC arrow** column shows which Theory of Change arc each anchor feeds.

A9 · Robustness — three-way OLS interactions

To verify the heterogeneity detected by the regression forest, we estimate three-way interactions with OLS. If the forest correctly identifies compound subgroups, the three-way coefficients should be significant for the outcomes with most heterogeneity.

3-way interaction	Attending school	Enrolled	HH internet
Girl × Rural × Indigenous	-0.056 (0.040)	-0.037 (0.039)	0.094 (0.086)
Girl × Rural × Poor	-0.005 (0.040)	-0.004 (0.038)	-0.097 (0.079)
Girl × Indigenous × Poor	0.054 (0.047)	0.063 (0.045)	0.152 (0.095)

Three-way interaction coefficients in decimal points (× 100 for pp). SE in parentheses. * p<0.10, ** p<0.05, *** p<0.01. Most three-way coefficients are not significant — the forest detects heterogeneity primarily through individual variables (rurality, department), not their compound interaction. **Interpretation:** exclusion accumulates additively more than multiplicatively.

A10 · Structural Equation Model (SEM) — proposed

Next step to integrate all anchors into a single model of the Theory of Change. We pre-register the structure here; estimation requires pilot + MICS Bolivia 2019 + costing data.

Measurement model (which observable indicators load on each latent construct):

$$\text{TeacherCap}_s = \lambda_1 \text{TrainHours}_s + \lambda_2 \text{DigCompEdu}_s + \varepsilon$$

$$\text{GirlSkills}_i = \lambda_3 \text{CompThink}_i + \lambda_4 \text{DigConfidence}_i + \varepsilon$$

$$\text{HomeEnv}_h = \lambda_5 \text{ParentEng}_h + \lambda_6 \text{GenderNorm}_h + \varepsilon$$

Structural model (Theory of Change arrows):

$$\text{ClassPractice}_s = \beta_1 \text{TeacherCap}_s + u_1$$

$$\text{GirlSkills}_i = \beta_2 \text{ClassPractice}_{s(i)} + \beta_3 \text{HomeEnv}_{h(i)} + u_2$$

$$\text{Attendance}_i = \beta_4 \text{GirlSkills}_i + \beta_5 \text{HomeEnv}_{h(i)} + u_3$$

$$\text{Progression}_i = \beta_6 \text{Attendance}_i + \beta_7 \text{GirlSkills}_i + u_4$$

A11 · Limitations

- EH measures **access to devices and internet**, not actual digital skills
- HH access **does not equal meaningful use** by the adolescent
- Disability sample is **small** — interpret with caution
- Data are **cross-sectional**: no temporal trajectory observed
- **Direct measurement at pilot level** is needed to validate and refine the projections

hgrueso@unicef.org